

Distributional Semantic Models

Part 3: Evaluation – is my DSM “good”?

Stephanie Evert¹

with Gabriella Lapesa², Alessandro Lenci³, and Marco Baroni⁴

¹Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

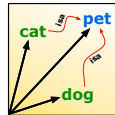
²Heinrich-Heine-Universität Düsseldorf & GESIS, Germany

³University of Pisa, Italy

⁴ICREA / Universitat Pompeu Fabra Barcelona, Italy

<http://wordspace.collocations.de/doku.php/course:start>

Copyright © 2009–2026 Evert, Lapesa, Lenci & Baroni | Licensed under CC-by-sa version 3.0



The problem

“The distributional hypothesis, as motivated by the works of Zellig Harris, is a strong methodological claim with a weak semantic foundation. It states that differences of meaning correlate with differences of distribution, but it neither specifies **what kind of distributional information we should look for**, nor **what kind of meaning differences it mediates**.” (Sahlgren 2008, 37)

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

The solution

Which kind of meaning differences does my DSM capture (if any)?

1. Parameter manipulation

- ▶ **What kind of information should we look for?**
- ▶ after Part 2, we are all experts and know how many different options we have!

2. Evaluation (task + datasets)

- ▶ **What kind of meaning differences do we capture?**
- ▶ for example, in an evaluation on a nearest neighbours task, our definition of “what a good neighbor is” is given by the dataset

3. Interpretation of the evaluation results

- ▶ crucial issue, often disregarded or oversimplified

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

- Multiple choice
- Prediction of similarity ratings
- Noun categorisation
- Analogy

Methodology for evaluation of DSM parameters

- An example (Bullinaria & Levy 2007, 2012)
- Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

- Tasks & parameters
- Evaluation results
- Summary & conclusion

Tasks & datasets

- **Tasks** are experimental setups to test DSM representations
 - **classification (multiple choice)**: given a target word, pick the “best” from a set of candidates (whatever best means)
 - **correlation**: do DSM similarities match scores that quantify semantic similarity/relatedness (ratings, reaction times)?
 - **categorisation**: do DSM similarities group words in a “reasonable” way?
- **Datasets** are the external “ground truth” and contribute the semantic “nuance” to the evaluation
 - collected ad-hoc for DSM evaluation or already existing, e.g. TOEFL, similarity ratings, experimental items from psycholinguistic experiments

{Task + Dataset} as operationalisation of a hypothesis, e.g.
DSM similarity as synonymy → multiple choice task + TOEFL

Tasks

Intrinsic vs. extrinsic tasks

- **Intrinsic evaluation**: the semantic representations produced by the DSM are evaluated *directly*
 - The DSM is the *only factor* responsible for the performance
- **Extrinsic evaluation**: the DSM representations are input to further tasks whose performance is then evaluated, e.g.
 - DSM vectors as input of a machine learning classifier
 - accuracy of the classifier
 - DSM vectors to improve a machine translation system
 - BLEU score of the MT

Datasets

Reminder: the many facets of DSM similarity

- **Attributional similarity** – two words sharing a large number of salient features (attributes)
 - synonymy (*car/automobile*)
 - hyperonymy (*car/vehicle*)
 - co-hyponymy (*car/van/truck*)
- **Semantic relatedness** (Budanitsky & Hirst 2006) – two words semantically associated without necessarily being similar
 - function (*car/drive*)
 - meronymy (*car/tyre*)
 - location (*car/road*)
 - attribute (*car/fast*)
- **Relational similarity** (Turney 2006) – similar relation between pairs of words (analogy)
 - *policeman:gun :: teacher:book*
 - *mason:stone :: carpenter:wood*
 - *traffic:street :: water:riverbed*

Datasets for intrinsic evaluation

of attributional similarity and semantic relatedness

- ▶ **Synonym identification**
 - ▶ TOEFL test (Landauer & Dumais 1997)
- ▶ **Modeling semantic similarity** judgments
 - ▶ RG norms (Rubenstein & Goodenough 1965)
 - ▶ WordSim-353 (Finkelstein *et al.* 2002)
 - ▶ MEN (Bruni *et al.* 2014), SimLex-999 (Hill *et al.* 2015)
- ▶ **Noun categorisation**
 - ▶ ESSLLI 2008 dataset
 - ▶ Almuhareb & Poesio (AP, Almuhareb 2006)
- ▶ **Semantic priming**
 - ▶ Hodgson dataset (Padó & Lapata 2007)
 - ▶ Semantic Priming Project (Hutchison *et al.* 2013)
- ▶ **Analogies & semantic relations** (intrinsic & extrinsic, ML)
 - ▶ Google (Mikolov *et al.* 2013b), BATS (Gladkova *et al.* 2016)
 - ▶ BLESS (Baroni & Lenci 2011), CogALex (Santus *et al.* 2016)

Give it a try ...

- ▶ The wordspace package contains pre-compiled DSM vectors
 - ▶ based on a large Web corpus (9 billion words)
 - ▶ L4/R4 surface span, log-transformed G^2 , SVD dim. red.
 - ▶ targets = lemma + POS code (e.g. *white_J*)
 - ▶ compatible with evaluation tasks included in package

```
library(wordspace)

M <- DSM_Vectors
nearest.neighbours(M, "walk_V")
  amble_V  stroll_V  traipse_V  potter_V  tramp_V
    19.4    21.8    21.8    22.6    22.9
  saunter_V wander_V  trudge_V leisurely_R saunter_N
    23.5    23.7    23.8    26.2    26.4
# you can also try white, apple and kindness
```

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

The TOEFL synonym task

- ▶ The TOEFL dataset (80 items)
 - ▶ Target: *show*
 - Candidates: *demonstrate*, *publish*, *repeat*, *postpone*
 - ▶ Target *costly*
 - Candidates: *beautiful*, *complicated*, *expensive*, *popular*
- ▶ DSMs and TOEFL
 1. take vectors of the target (**t**) and of the candidates (**c**₁ ... **c**_n)
 2. measure the distance between **t** and **c**_i, with 1 ≤ i ≤ n
 3. select **c**_i with the shortest distance in space from **t**

```
# ask your course instructor for non-public data package
> library(wordspaceEval)
> head(TOEFL80)
```

Humans vs. machines on the TOEFL task

- ▶ Average foreign test taker: 64.5%
- ▶ Macquarie University staff (Rapp 2004):
 - ▶ Average of 5 non-natives: 86.75%
 - ▶ Average of 5 natives: **97.75%**
- ▶ Distributional semantic models (ACL wiki)
 - ▶ Term-Document: LSA (Landauer & Dumais 1997): 64.4%
 - ▶ Dependency-filtered (Padó & Lapata 2007): 73.0%
 - ▶ Dependency-typed (Baroni & Lenci 2010): 76.9%
 - ▶ Term-term Bullinaria & Levy (2012): **100.0%** (after aggressive parameter optimisation)

And you?

```
> eval.multiple.choice(TOEFL80, M)
```

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

Semantic similarity judgments

RG65

65 pairs, rated from 0 to 4

gem – *jewel*: 3.94

grin – *smile*: 3.46

fruit – *furnace*: 0.05

WordSim353

353 pairs, rated from 1 to 10

announcement – *news*: 7.56

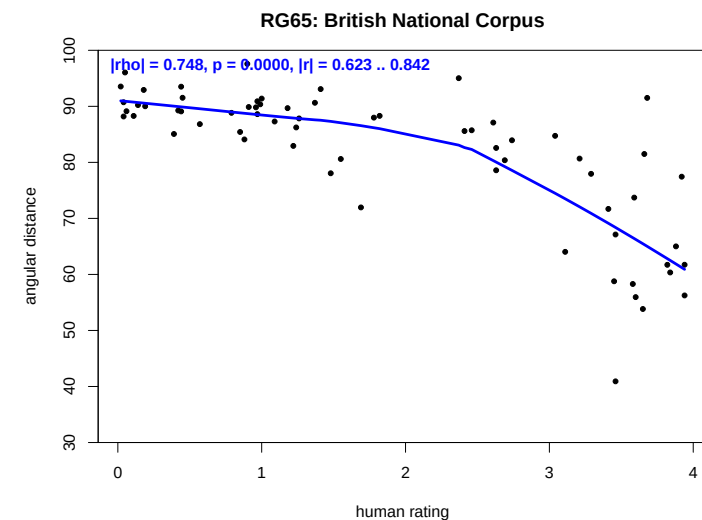
weapon – *secret*: 6.06

travel – *activity*: 5.00

- ▶ DSMs vs. Ratings: operationalisation
 1. for each test pair (w_1, w_2), take vectors $\mathbf{w}_1$ and $\mathbf{w}_2$
 2. measure the distance (e.g. cosine) between $\mathbf{w}_1$ and $\mathbf{w}_2$
 3. measure (Pearson) correlation between vector distances and average similarity judgments (Padó & Lapata 2007)

```
> RG65[seq(0,65,5), ]
> head(WordSim353) # extension of Rubenstein-Goodenough
```

Semantic similarity judgments: example



Semantic similarity judgments: results

Results on RG65 task (Pearson):

- ▶ Dependency-filtered, BNC: Padó and Lapata (2007): 0.62
- ▶ Dependency-filtered, Web data (Herdağdelen *et al.* 2009)
 - ▶ without SVD reduction: 0.69
 - ▶ with SVD reduction: 0.80
- ▶ Dependency typed (Baroni & Lenci 2010): 0.82
- ▶ Term-term + some magic (Hassan & Mihalcea 2011): 0.86

And you?

```
> eval.similarity.correlation(RG65, M, convert=FALSE)
      rho p.value missing      r r.lower r.upper
RG65 0.687 2.61e-10      0 0.678 0.52 0.791
> plot(eval.similarity.correlation( # cosine similarity
      RG65, M, convert=FALSE, details=TRUE))
```

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

Noun categorisation

- ▶ In **categorisation tasks**, subjects are typically asked to assign experimental items – objects, images, words – to a given category or group items belonging to the same category
 - ▶ categorisation requires an understanding of the relationship between the items in a category
- ▶ Categorisation is a basic cognitive operation and a precondition of many further semantic tasks
 - ▶ **inference**
 - ★ if X is a CAR then X is a VEHICLE
 - ▶ **compositionality**
 - ★ $\lambda y : \text{FOOD } \lambda x : \text{ANIMATE } [\text{eat}(x, y)]$
- ▶ “Chicken-and-egg” problem for relationship of categorisation and similarity (cf. Goodman 1972, Medin et al. 1993)

Noun categorisation: datasets

ESSLLI08 (on focus today)

44 nouns, 6 classes

potato $\Rightarrow$ GREEN

hammer $\Rightarrow$ TOOL

car $\Rightarrow$ VEHICLE

peacock $\Rightarrow$ BIRD

BATTIG set

82 nouns, 10 classes

chicken $\Rightarrow$ BIRD

bear $\Rightarrow$ LAND_MAMMAL

pot $\Rightarrow$ KITCHENWARE

oak $\Rightarrow$ TREE

Almuhareb & Poesio

402 nouns, 21 classes

day $\Rightarrow$ TIME

kiwi $\Rightarrow$ FRUIT

kitten $\Rightarrow$ ANIMAL

volleyball $\Rightarrow$ GAME

MITCHELL set

60 nouns, 12 classes

ant $\Rightarrow$ INSECT

carrot $\Rightarrow$ VEGETABLE

train $\Rightarrow$ VEHICLE

cat $\Rightarrow$ ANIMAL

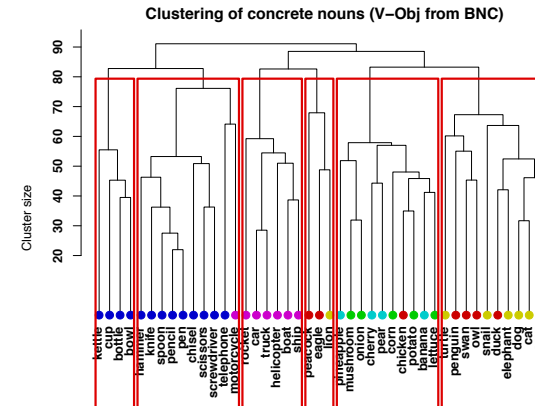
Noun categorisation: the ESSLI 2008 dataset

Dataset of 44 concrete nouns (ESSLI 2008 Shared Task)

- ▶ 24 natural entities
 - ▶ 15 animals: 7 birds (*eagle*), 8 ground animals (*lion*)
 - ▶ 9 plants: 4 fruits (*banana*), 5 greens (*onion*)
- ▶ 20 artifacts
 - ▶ 13 tools (*hammer*), 7 vehicles (*car*)
- ▶ DSMs operationalise categorisation as a **clustering task**
 1. for each noun w_i in the dataset, take its vector $\mathbf{w}_i$
 2. use a **clustering method** to group similar vectors $\mathbf{w}_i$
 3. evaluate whether clusters correspond to gold-standard semantic classes (purity, entropy, ...)

```
> ESSLI08_Nouns[seq(1,40,5), ]
```

Noun categorisation: example



- ▶ majority labels: **tools**, **tools**, **vehicles**, **birds**, **greens**, **animals**
- ▶ correct: 4/4, 9/10, 6/6, 2/3, 5/10, 7/11
- ▶ purity = 33 correct out of 44 = 75.0%

ESSLI 2008 shared task

- ▶ Clustering experiments with CLUTO (Karypis 2003)
 - ▶ 6-way (birds, ground animals, fruits, greens, tools and vehicles), 3-way (animals, plants and artifacts) and 2-way (natural and artificial entities) clusterings
- ▶ Evaluation scores:
 - ▶ **purity** – degree to which a cluster contains words from one class only (**best = 1**)
 - ▶ **entropy** – whether words from different classes are represented in the same cluster (**best = 0**)
 - ▶ **global score** across the three clustering experiments

$$\sum_{i=1}^3 \text{Purity}_i - \sum_{i=1}^3 \text{Entropy}_i$$

ESSLI 2008 shared task

model	6-way		3-way		2-way		global
	P	E	P	E	P	E	
Pattern-based (Katrénko)	89	13	100	0	80	59	197
Term-term (Peirsman)	82	23	84	34	86	55	140
dep-typed (DM)	77	24	79	38	59	97	56
dep-filtered (DM)	80	28	75	51	61	95	42
window (DM)	75	27	68	51	68	89	44

Katrénko, Peirsman: ESSLI 2008 Shared Task
DM: Baroni & Lenci (2009)

And you?

```
> eval.clustering(ESSLI08_Nouns, M) # uses PAM clustering
```

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

Intrinsic evaluation on word pairs: Analogy

Mikolov et al. (2013b,a); Gladkova et al. (2016)

► Task: solve analogy problems such as

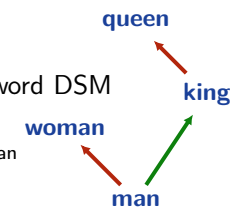
- *man* : *woman* :: *king* : *queen*
- *France* : *Paris* :: *Bulgaria* : *Sofia*
- *learn* : *learned* :: *go* : *went*
- *dog* : *animal* :: *strawberry* : *fruit*

► Approach 1: build DSM on word pairs as targets

$$\min_x d(\mathbf{v}_{\text{man:woman}}, \mathbf{v}_{\text{king:x}})$$

► Approach 2: use vector operations in single-word DSM

$$\mathbf{v}_{\text{queen}} \approx \mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}}$$



The Google analogy task

Mikolov et al. (2013b,a)

Table 1: Examples of five types of semantic and nine types of syntactic questions in the Semantic-Syntactic Word Relationship test set.

Type of relationship	Word Pair 1		Word Pair 2	
Common capital city	Athens	Greece	Oslo	Norway
All capital cities	Astana	Kazakhstan	Harare	Zimbabwe
Currency	Angola	kwanza	Iran	rial
City-in-state	Chicago	Illinois	Stockton	California
Man-Woman	brother	sister	grandson	granddaughter
Adjective to adverb	apparent	apparently	rapid	rapidly
Opposite	possibly	impossibly	ethical	unethical
Comparative	great	greater	tough	tougher
Superlative	easy	easiest	lucky	luckiest
Present Participle	think	thinking	read	reading
Nationality adjective	Switzerland	Swiss	Cambodia	Cambodian
Past tense	walking	walked	swimming	swam
Plural nouns	mouse	mice	dollar	dollars
Plural verbs	work	works	speak	speaks

(Mikolov et al. 2013b, Tab. 1)

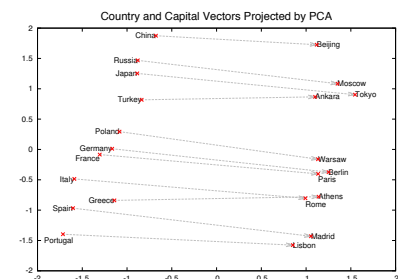
The Google analogy task

Mikolov et al. (2013b,a)

► Mikolov et al. (2013b,a)

claim that their neural embeddings are good at solving analogy tasks

► Semantic features encoded in linear subdimensions



(Mikolov et al. 2013a, Fig. 2)

model	syntactic	semantic	
word2vec	64%	55%	(Mikolov et al. 2013b)
DSM	43%	60%	(Baroni et al. 2014)
FastText	82%	87%	(Mikolov et al. 2018)

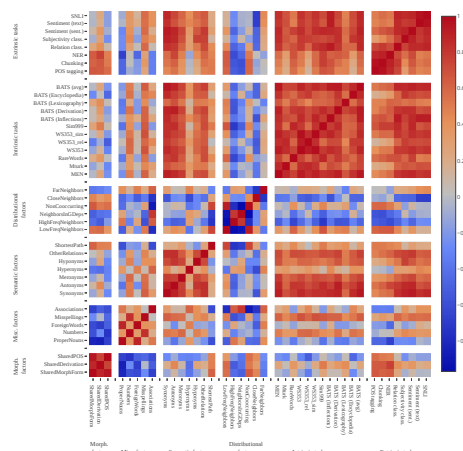
Scaling up: Linguistic Diagnostics

Linguistic Diagnostics

(Rogers *et al.* 2018)

automates classification of nearest neighbours based on various on-line dictionaries and semantic networks

- correlation analysis between various groups of diagnostics and evaluation tasks
- only for English so far



<https://github.com/annargr/ldt>

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

DSM evaluation in published studies

- One model, many tasks (Padó & Lapata 2007; Baroni & Lenci 2010; Pennington *et al.* 2014)
 - A novel DSM is proposed, with specific features & parameters
 - This DSM is tested on a range of different tasks (e.g. TOEFL, priming, semantic clustering)
- Incremental tuning of parameters (Bullinaria & Levy 2007, 2012; Kiela & Clark 2014; Polajnar & Clark 2014)
 - Several parameters (e.g., scoring measure, distance metric, dimensionality reduction)
 - Many tasks (e.g. TOEFL, semantic & syntactic clustering)
 - Varying granularity of parameter settings
 - One parameter (sometimes two) varied at a time, with all other parameters set to fixed values or optimized for each setting
 - Optimal parameter values are determined sequentially

Bullinaria & Levy (2007, 2012)

- One of the first systematic evaluation studies
- Test influence of many standard parameter settings
 - frequency weighting + distance measure
 - co-occurrence window, structured *vs.* unstructured
 - corpus type & size, number of feature dimensions
 - dimensionality reduction (SVD), number of latent dimension
- In five different evaluation tasks
 - TOEFL
 - distance comparison: related word *vs.* 10 random words
 - semantic categorisation: nearest-centroid classifier
 - syntactic categorisation (2007)
 - semantic clustering of nouns (2012)
- Novel parameters
 - skipping of first latent dimensions (with highest variance)
 - Caron's (2001) *P*: power-scaling of singular values

DSM Tutorial – Part 3

- Methodology for evaluation of DSM parameters
 - An example (Bullinaria & Levy 2007, 2012)
 - Bullinaria & Levy (2007, 2012)

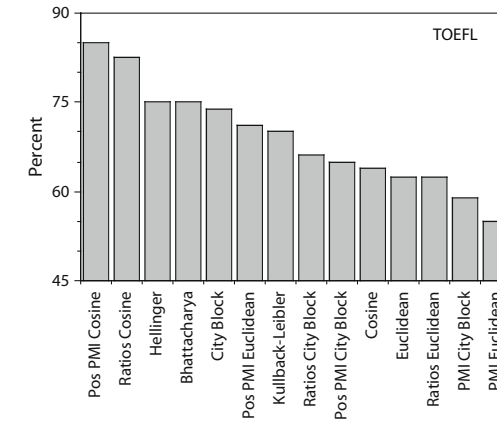
Bullinaria & Levy (2007, 2012)

- One of the first systematic evaluation studies
- Test influence of many standard parameter settings
 - frequency weighting + distance measure
 - co-occurrence window, structured vs. unstructured
 - corpus type & size, number of feature dimensions
 - dimensionality reduction (SVD), number of latent dimensions
- In face different evaluation tasks
 - TOEFL
 - distance comparison: related word vs. 10 random words
 - semantic categorization: nearest-neighbor classifier
 - semantic categorization (2007)
 - semantic clustering of nouns (2012)
- Noun parameters
 - clipping of first latent dimension (with highest variance)
 - Caron's (2001) P -power-ranking of singular values

- Note that B&L do not test all commonly used parameter settings, e.g. log-likelihood or log frequency as feature weighting.
- They also do not seem to normalize row vectors properly (frequency vectors are scaled to probabilities, i.e. $\|x\|_1 = 1$), which might explain the poor results from Euclidean distance.
- Noun clustering uses Mitchell *et al.* (2008) dataset of 60 nouns from 12 categories, and CLUTO direct clustering with standard settings.
- Incremental approach also across the two studies: B&L (2012) builds on and extends results of B&L (2007).
- Note that B&L's power scaling notation differs from Caron (2001): $P = 1 + p/2$

TOEFL results: feature weighting & distance measure

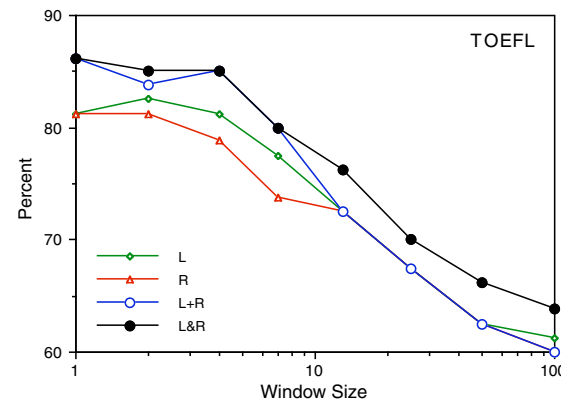
(Bullinaria & Levy 2007, p. 516, Fig. 1)



British National Corpus (BNC). Vectors not L2-normalized (frequency is L1-normalized). All other parameters optimized for each setting.

TOEFL results: size & type of co-occurrence window

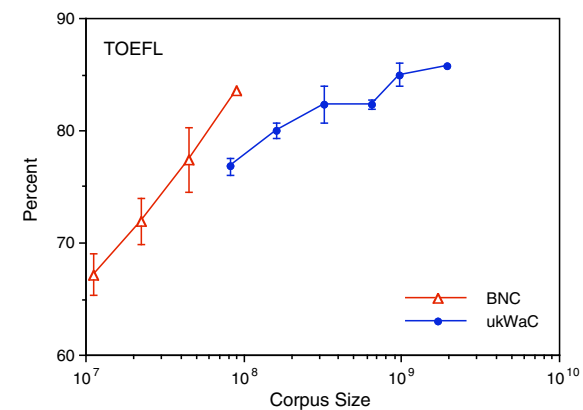
(Bullinaria & Levy 2012, p. 893, Fig. 1)



ukWaC Web corpus. Positive PMI + cosine (Bullinaria & Levy 2007). Number of feature dimensions optimized for each window size & task. No dimensionality reduction. L&R = structured surface context (left/right).

TOEFL results: corpus type & size

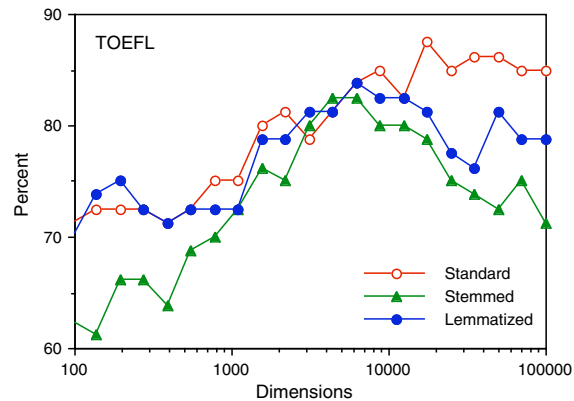
(Bullinaria & Levy 2012, p. 894, Fig. 2)



L+R context of size 1. Average + standard error over equally-sized corpus slices. Other parameter settings unclear.

TOEFL results: feature dimensions & pre-processing

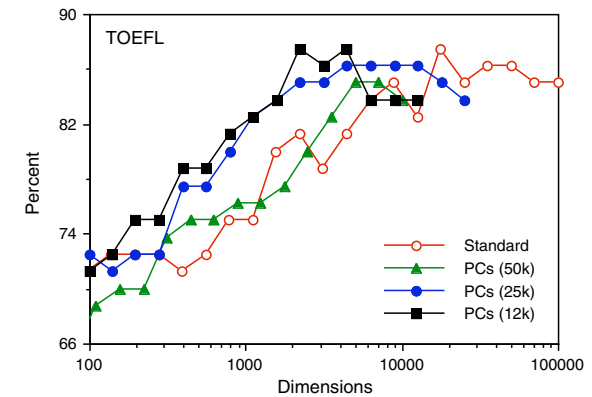
(Bullinaria & Levy 2012, p. 895, Fig. 4)



ukWaC corpus. L+R context of size 1. Other parameters presumably as above.

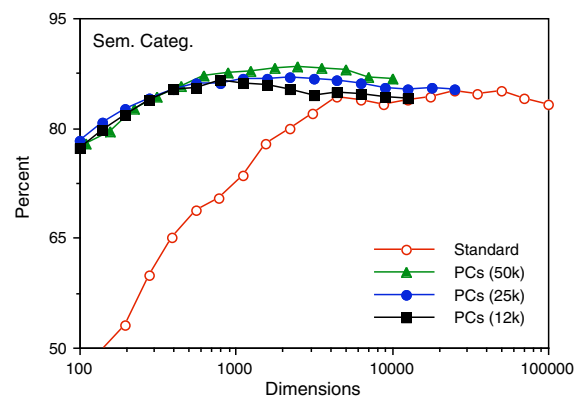
TOEFL results: dimensionality reduction

(Bullinaria & Levy 2012, p. 898, Fig. 5)

ukWaC corpus. Positive PMI + cosine. Standard = no dimensionality reduction.
Other: number of latent dimensions for 12k, 25k and 50k original feature dimensions.

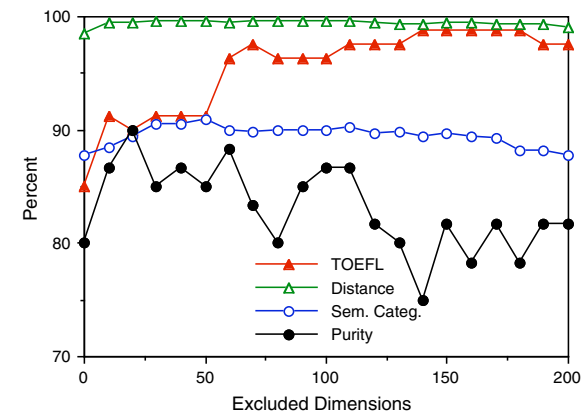
Semantic categorisation: dimensionality reduction

(Bullinaria & Levy 2012, p. 898, Fig. 5)

ukWaC corpus. Positive PMI + cosine. Standard = no dimensionality reduction.
Other: number of latent dimensions for 12k, 25k and 50k original feature dimensions.

Combined results: skipping first latent dimensions

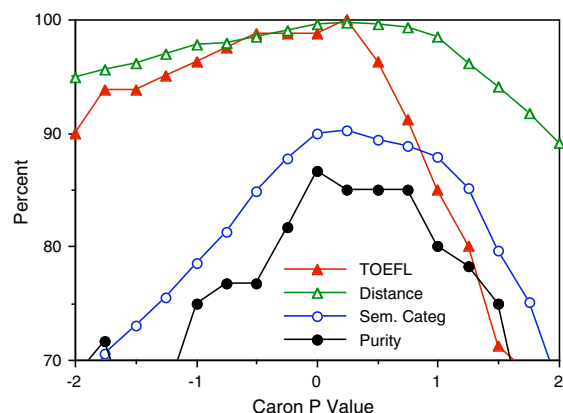
(Bullinaria & Levy 2012, p. 900, Fig. 7)



ukWaC corpus with standard settings. 50k feature dimensions reduced to 5000 latent dimensions.

TOEFL results: power scaling (Caron's P)

(Bullinaria & Levy 2012, p. 900, Fig. 7)



ukWaC corpus with standard settings. 50k feature dimensions reduced to 5000 latent dimensions. Neutral value is $P = 1$.

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

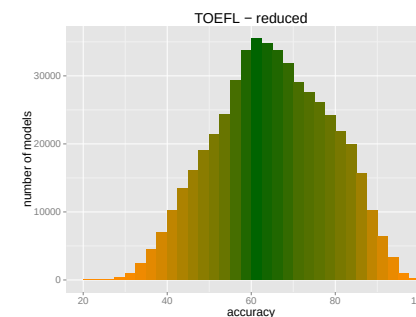
Making sense of evaluation results

Interpreting performance vs. picking the best run

- One model, many tasks** (Padó & Lapata 2007; Baroni & Lenci 2010; Pennington *et al.* 2014)
 - Novel DSM, one (or very few) settings tested on many tasks
 - Problem: not suitable for the exploration of a large parameter set, very limited coverage of interactions
- Incremental tuning** (Bullinaria & Levy 2007, 2012; Kiela & Clark 2014; Polajnar & Clark 2014)
 - Set parameter a , then b , then c
 - Problem: order dependent, little coverage of interactions
- Test all combinations** (Baroni *et al.* 2014; Levy *et al.* 2015; Lapesa & Evert 2014)
 - Many tasks, many parameters, all combinations
 - Problem: many runs, **interpreting results is a challenge**

Lots of variation to make sense of...

TOEFL: 504,000 (!!!) runs (Lapesa & Evert 2014)



We need an interpretation methodology that:

- ... is able to identify robust trends, avoiding overfitting
- ... is able to capture parameter interactions

Linear regression to the rescue

- Attempts to predict the values of a “dependent” variable from one or more “independent” variables and their combinations
- Is used to understand **which independent variables are closely related to the dependent variable**, and to **explore the forms of these relationships**

Example

Dependent variable: income

Independent variables: gender, age, ethnicity, education level, first letter of the surname (hopefully not significant)

How to interpret the evaluation results?

Our proposal: linear regression

We use linear models to analyze the influence of different DSM parameters and their combinations on DSM performance

- dependent variable = **performance**
(accuracy, correlation coefficient, purity)
- independent variables = model **parameters**
(e.g. source corpus, window size, association score)

Motivation

We want to understand which of the parameters are related to the dependent variable, i.e. we want to find the parameters whose manipulation has the strongest effect on DSM performance.

How to interpret the evaluation results?

Our proposal: linear regression

$$\text{model performance} = \beta_0 + \beta_1(p_1) + \beta_2(p_2) + \beta_3(p_1, p_2) + \dots + \epsilon$$

1. **Adjusted R^2 :** proportion of variance explained by the model
~> How well do we predict performance?
2. **Feature ablation:** proportion of variance explained by a parameter together with all its interactions
~> Which parameters affect performance the most?
3. **Model predictions:** visualisation of predicted performance
~> What are the best parameter values?

How well do we predict performance?

A concrete example: TOEFL, SVD (504k data points)

$$\text{accuracy} \sim \text{corpus} * \text{window} * \text{score} * \text{transformation} * \text{metric} * \text{rel.index} * \text{n.dim} * \text{dim.skip}$$

corpus	window	score	transformation	metric	n.dim	dim.skip	rel.index	accuracy
wacky	8	t-score	none	manhattan	700	0	dist	71.25
bnc	16	z-score	root	cosine	100	100	rank	75.00
wacky	16	MI	log	cosine	100	50	dist	77.50
bnc	8	frequency	none	cosine	900	50	rank	75.00
ukwac	16	MI	none	cosine	500	100	rank	81.25
bnc	8	tf.idf	root	cosine	300	100	rank	75.00
bnc	16	tf.idf	root	manhattan	300	100	dist	51.25
ukwac	2	tf.idf	log	manhattan	300	50	rank	53.75
ukwac	1	simple-ll	log	manhattan	500	100	dist	85.00

Model fit: Adj. R^2

basic 43%

& SVD +24%

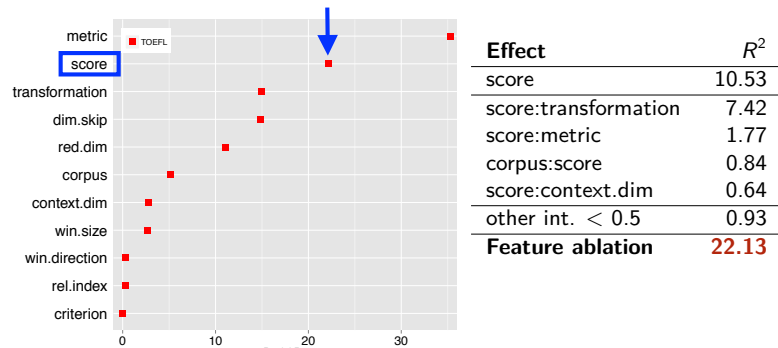
& 2-way +22%

Total: 87%

Assumption: a good linear model acts as a “smoothing” algorithm which filters away random noise & captures robust trends.

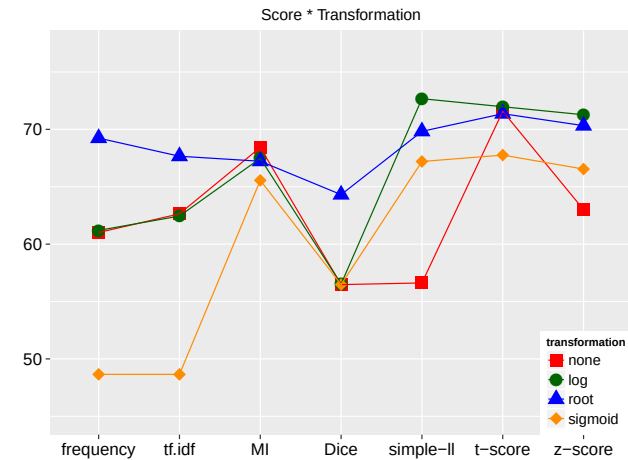
Which parameters affect performance the most?

Feature ablation: parameters and interactions on TOEFL



Which parameters affect performance the most?

Interaction of score and transformation: effect plot



So, are there general trends? (Lapesa & Evert 2014)

Datasets: TOEFL, RG65, WordSim353, ESSLLI08 (and 3 other clust. datasets)

- ▶ Most explanatory parameters: similar across tasks/datasets
 - ▶ simple-ll + logarithmic transformation, cosine similarity
- ▶ Parameters that show variation: **the amount and nature** of shared context
 - ▶ Context window: 4 is a good compromise solution
 - ▶ SVD: always helps, and skipping the first dimensions (but not too many) generally helps
- ▶ Neighbor rank (almost) always better than distance
- ▶ Syntax (almost) never helps :((Lapesa & Evert 2017)

Appendix A (very) large-scale evaluation study

(Lapesa & Evert 2014)

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

Tasks

1. Classification

- ▶ **TOEFL80**: multiple-choice classification task (Landauer & Dumais 1997)

2. Correlation to Similarity Ratings

- ▶ **RG65**: 65 noun pairs (Rubenstein & Goodenough 1965)
- ▶ **WordSim353**: 351 noun pairs (Finkelstein *et al.* 2002)

3. Semantic Clustering

- ▶ **Battig82**: 82 nouns, 10 classes (Van Overschelde *et al.* 2004)
- ▶ **AP402**: 402 nouns, 21 classes (Almuhareb 2006)
- ▶ **ESSLLI08_Nouns**: 44 nouns, 6 classes
- ▶ **Mitchell**: 60 nouns, 12 classes (Mitchell *et al.* 2008)

Distributional models: general features

- ▶ **Term-term** distributional semantic models (bag-of-words)
- ▶ **Target** terms (rows)
 - ▶ vocabulary from Distributional Memory (Baroni & Lenci 2010) + terms from evaluation datasets
 - ▶ 27522 lemma types
- ▶ **Feature** terms (columns)
 - ▶ filtered by part-of-speech (nouns, verbs, adjectives, adverbs)
 - ▶ further context selection determined by two model parameters

Distributional models were compiled and evaluated using the IMS **Corpus Workbench**¹, the **UCS toolkit**² and the **wordspace package** for R.

¹<http://cwb.sf.net/>

²<http://www.collocations.de/software.html>

Evaluated parameters

Building the co-occurrence matrix

1. **Source corpus**: BNC, Wackypedia, UkWac

Our source corpora – standard choices in distributional semantics – differ in both size and quality. Is there a quantity/quality trade-off?

2. **Window** (= surface span)

- ▶ **Direction**: directed (= structured), undirected
- ▶ **Size**: 1, 2, 4, 8, 16 words

We expect those parameters to be crucial as they determine the granularity (direction) and amount (size) of shared context involved in the computation of similarity.

Evaluated parameters

Selecting dimensions from the co-occurrence matrix

3. Feature selection:

- ▶ **Criterion:** frequency, number of non-zero entries
- ▶ **Threshold:** top n dimensions ($n = 5000, 10000, 20000, 50000, 100000$)

How many context dimensions (words) are needed for DSMs to perform well in specific tasks? Are too many context dimensions detrimental? What is the best selection criterion?

Evaluated parameters

Dimensionality reduction

6. Dimensionality reduction with randomized SVD:

- ▶ **number of reduced dimensions:** 100, 300, 500, 700, 900
- ▶ **number of skipped dimensions:** 0, 50, 100

Dimensionality reduction is expected to improve semantic representation and make computations more efficient. How does SVD interact with the other parameters? Bullinaria & Levy (2012) report improvements in some tasks (e.g. TOEFL) when the first latent dimensions (with highest variance) are discarded. Does this result generalize to our tasks/datasets?

Evaluated parameters

Weighting and scaling co-occurrence counts

4. Feature scoring: frequency, simple-II, MI, Dice, t-score, z-score, tf.idf

Association measures represent an interpretation of co-occurrence frequency, and they emphasize different types of collocations (Evert 2008). Does this have an effect on DSM performance?

5. Transformation: none, logarithmic, square root, sigmoid

Transformations reduce the skewness of feature scores.

Evaluated parameters

Computation and usage of distances

7. Distance metric: cosine (angular distance), manhattan

Both are symmetric, while cognitive processes are often asymmetric

8. Index of distributional relatedness

- ▶ **distance:** $\text{dist}(a, b)$
- ▶ **neighbor rank**, calculated differently for different tasks:
 - ★ TOEFL: backward rank, i.e. $\text{rank}(b, a)$
 - ★ Ratings and Clustering: average of logarithmic forward and backward rank, i.e. $(\log \text{rank}(a, b) + \log \text{rank}(b, a))/2$

This parameter allows us to account for asymmetries: $\text{rank}(b, a)$ is different from $\text{rank}(a, b)$. While cognitively plausible, neighbor rank is computationally expensive: does it improve the performance of DSMs?

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

TOEFL multiple-choice classification task

Introducing the task

A collection of 80 multiple-choice questions from a synonym task in the Test Of English as a Foreign Language (TOEFL)

TOEFL dataset

Target: *consume* – Choices: *breed*, *catch*, *eat*, *supply*

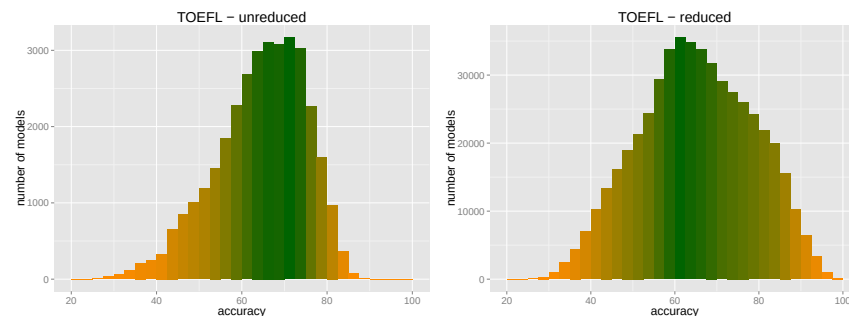
Target: *constant* – Choices: *accidental*, *continuing*, *instant*, *rapid*

Target: *concise* – Choices: *free*, *positive*, *powerful*, *succinct*

- ▶ A **classification** task
- ▶ If DSMs capture synonymy relations, we expect that the distance between the target and the correct choice will be smaller than to the wrong choices
- ▶ Performance: **% accuracy**

TOEFL task: performance

Unreduced versus Reduced Experiments

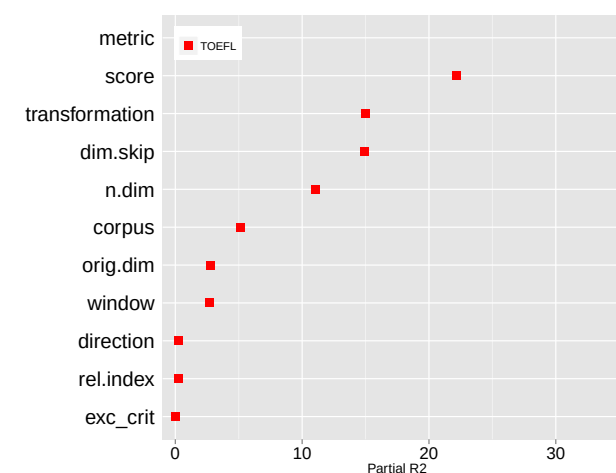


Min: 25 ; Max: 87.5 ; Mean: 63.9

Min: 18.7; Max: 97.4; Mean: 64.4

TOEFL task: parameters and explained variance

Reduced setting: feature Ablation (model R^2 : 89%)



TOEFL task: interactions

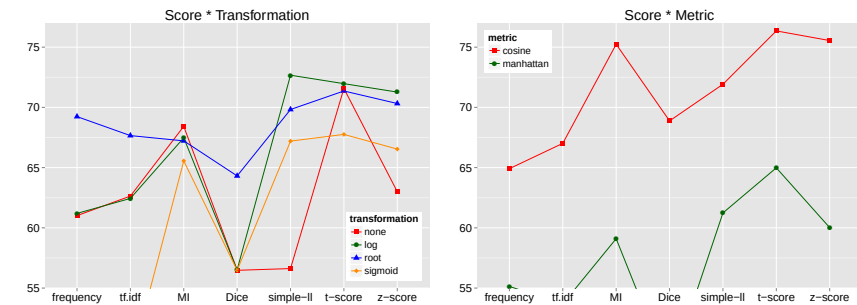
Reduced setting ($R^2 > 0.5$)

Interaction	Df	R^2
score:transformation	18	7.42
metric:dim.skip	2	4.44
score:metric	6	1.77
metric:orig.dim	4	0.98
window:transformation	12	0.91
corpus:score	12	0.84
score:orig.dim	24	0.64
metric:n.dim	4	0.63

TOEFL task: interactions, R^2

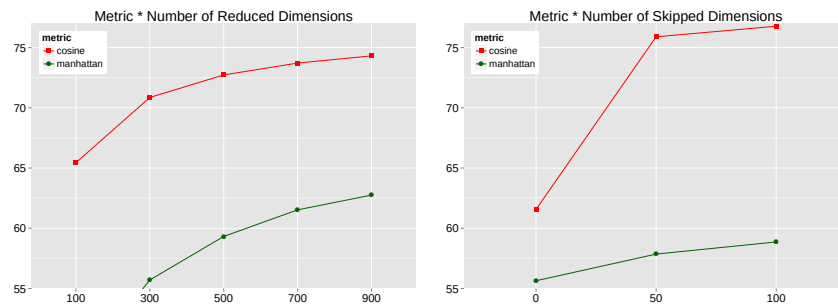
TOEFL task: Metric, Score, Transformation

Partial effect displays (Fox 2003)



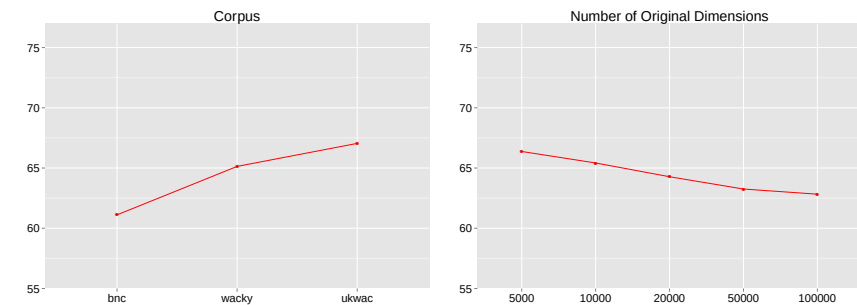
TOEFL task: Dimensionality Reduction

Partial effect displays (Fox 2003)



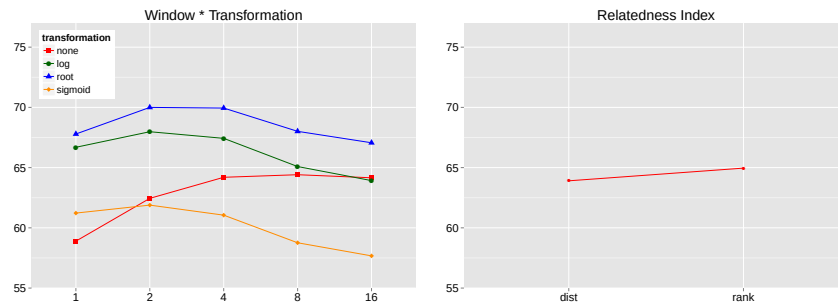
TOEFL task: Corpus and Number of Feature Dimensions

Partial effect displays (Fox 2003)



TOEFL task: Window and Relatedness Index

Partial effect displays (Fox 2003)



TOEFL task: summary

TOEFL: best setting

- ▶ Corpus: ukWac
- ▶ Window: undirected, 2 words
- ▶ Feature selection: top 5000/10000 dimensions, based on frequency
- ▶ Score * Transformation: simple-II * log
- ▶ Dimensionality Reduction: 900 latent dimensions, skipping the first 100
- ▶ Distance Metric: cosine
- ▶ Index of Distributional Relatedness: neighbor rank

DSMs and similarity ratings

Introducing the task

RG65

65 pairs, rated from 0 to 4

gem – *jewel*: 3.94

grin – *smile*: 3.46

fruit – *furnace*: 0.05

WordSim353

353 pairs, rated from 1 to 10

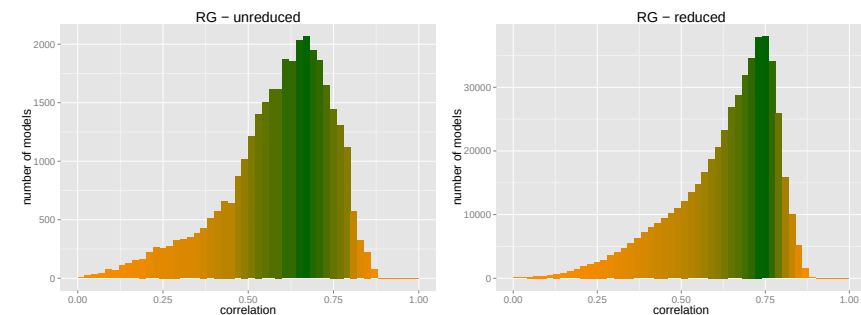
announcement – *news*: 7.56

weapon – *secret*: 6.06

travel – *activity*: 5.00

- ▶ A **prediction task**
- ▶ If distributional representation are close to speakers' conceptual representations, we expect to find some **correlation** between distance in the semantic space and speaker's judgments concerning semantic similarity
- ▶ Performance: **Pearson correlation r**

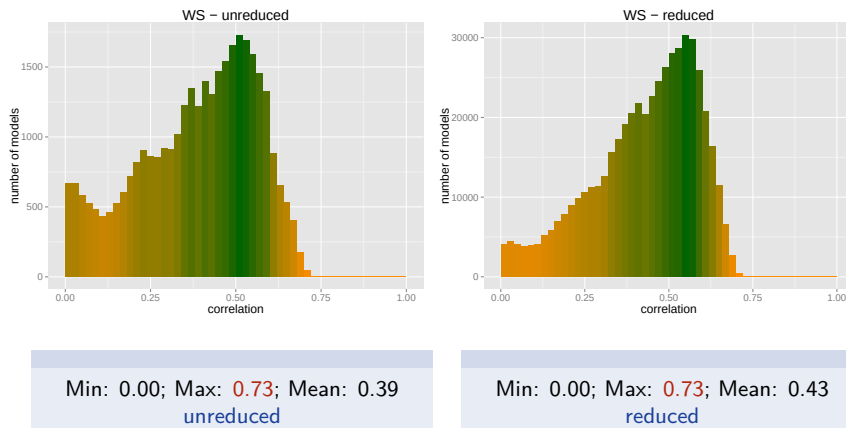
Similarity ratings: performance on RG65



Min: 0.01; Max: 0.88; Mean 0.59
unreduced

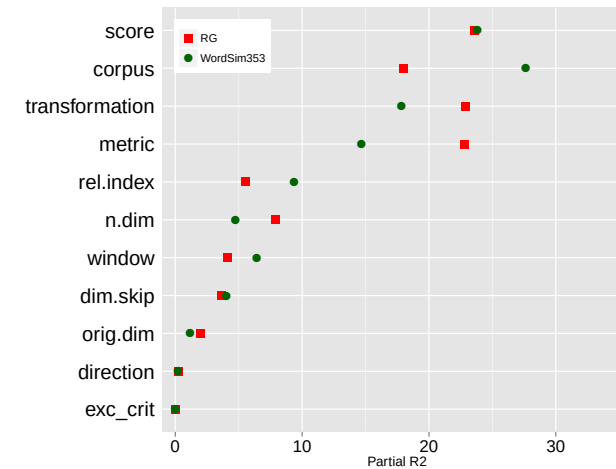
Min: 0.00; Max: 0.89; Mean: 0.63
reduced

Similarity ratings: performance on WordSim353



Similarity ratings: parameters and explained variance

Reduced setting: feature ablation (full model R^2 : RG65 86%; WS353 90%)



Similarity ratings: interactions

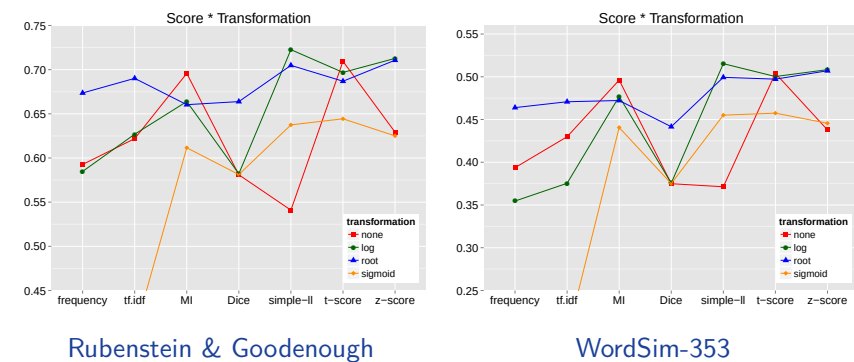
Reduced setting ($R^2 > 0.5$)

Interaction	Df	RG65	WordSim353
score:transf	18	10.28	8.66
metric:n.dim	4	2.18	1.42
window:transf	12	1.43	1.01
corpus:metric	2	1.83	0.51
score:metric	6	1.91	0.59
metric:orig.dim	4	1.08	0.62
corpus:score	12	0.77	0.82
window:score	24	0.77	0.69
score:dim.skip	12	0.58	0.85

Similarity ratings: interactions, R^2

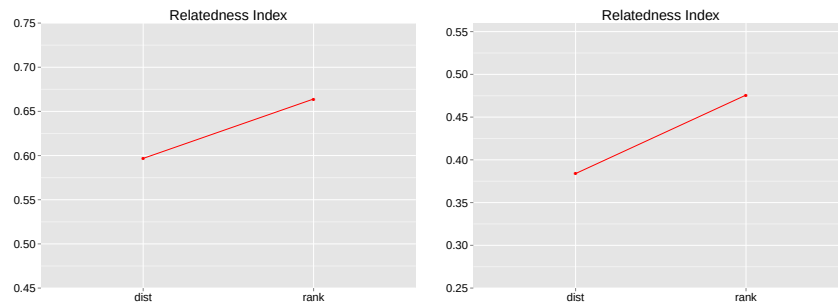
Similarity ratings: Score, Transformation

Partial effect displays (Fox 2003)



Similarity ratings: Relatedness Index

Partial effect displays (Fox 2003)

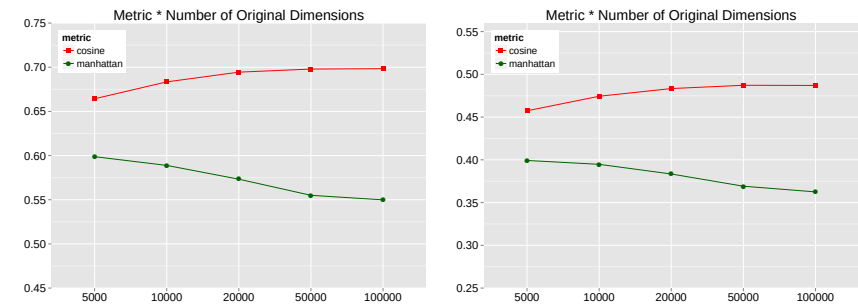


Rubenstein & Goodenough

WordSim-353

Similarity ratings: Metric, Number of Feature Dimensions

Partial effect displays (Fox 2003)

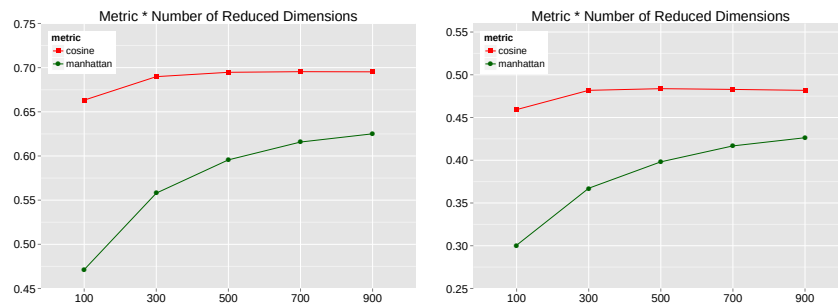


Rubenstein & Goodenough

WordSim-353

Similarity ratings: Number of Latent Dimensions

Partial effect displays (Fox 2003)

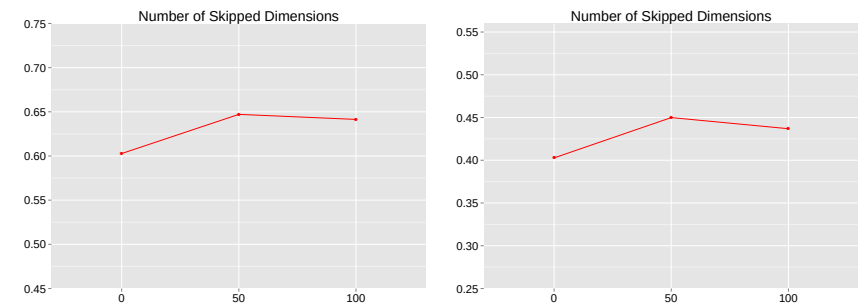


Rubenstein & Goodenough

WordSim-353

Similarity ratings: Number of Skipped Dimensions

Partial effect displays (Fox 2003)

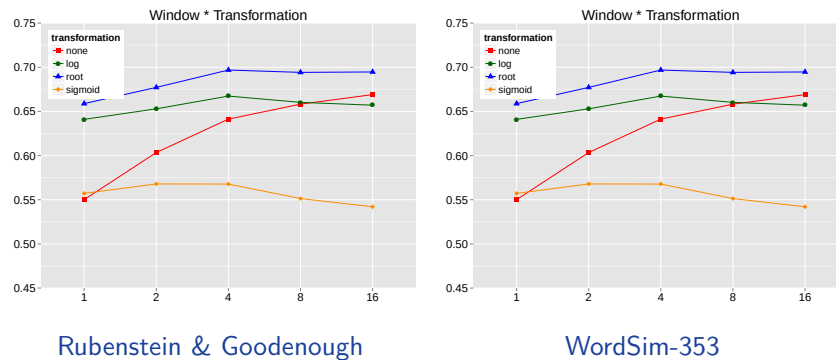


Rubenstein & Goodenough

WordSim-353

Similarity ratings: Window Size, Transformation

Partial effect displays (Fox 2003)



Summing up: Ratings

Ratings: best setting

- ▶ Corpus: wacky
- ▶ Window: undirected, 4 words
- ▶ Feature selection: top 20000/50000 dimensions, based on frequency
- ▶ Score * Transformation: simple-II * log
- ▶ Dimensionality Reduction: 300 latent dimensions, skipping the first 50
- ▶ Distance Metric: cosine
- ▶ Index of Distributional Relatedness: neighbor rank

DSMs and semantic clustering

Introducing the task

Almuhareb & Poesio

402 nouns, 21 classes

day ⇒ TIME

kiwi ⇒ FRUIT

kitten ⇒ ANIMAL

volleyball ⇒ GAME

ESSLLI categorization task

44 nouns, 6 classes

potato ⇒ GREEN

hammer ⇒ TOOL

car ⇒ VEHICLE

peacock ⇒ BIRD

BATTIG set

82 nouns, 10 classes

chicken ⇒ BIRD

bear ⇒ LAND_MAMMAL

pot ⇒ KITCHENWARE

oak ⇒ TREE

MITCHELL set

60 nouns, 12 classes

ant ⇒ INSECT

carrot ⇒ VEGETABLE

train ⇒ VEHICLE

cat ⇒ ANIMAL

DSMs and semantic clustering

Introducing the task

- ▶ A **categorization** task
- ▶ If distributional representations approximate human conceptual representations, we expect word categorization based on distributional features to produce concept clusters similar to those in the gold standard datasets
- ▶ Performance: **cluster purity**
 - ▶ classification accuracy for optimal cluster labelling
 - ▶ percentage of nouns that belong to the majority category within their cluster
- ▶ **Partitioning around medoids** (Kaufman & Rousseeuw 1990)
 - ▶ implemented as `pam()` in R standard library
 - ▶ direct comparison → equal to or even better than CLUTO
 - ▶ works with arbitrary dissimilarity matrix

Semantic clustering: performance

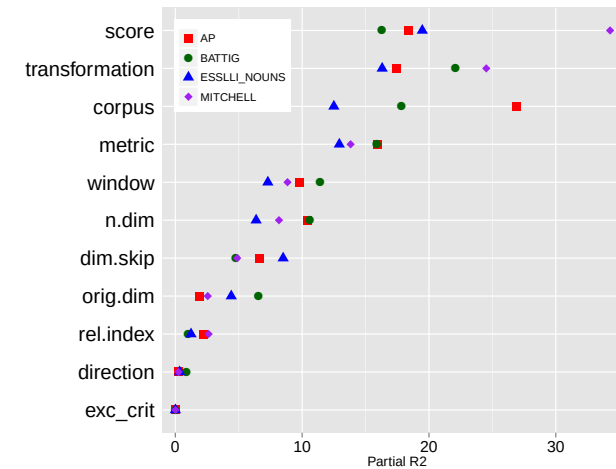
Overview: unreduced versus reduced experiments

Dataset	Unreduced			Reduced		
	Min	Max	Mean	Min	Max	Mean
AP	0.15	0.73	0.56	0.13	0.76	0.54
BATTIG	0.28	0.99	0.77	0.23	0.99	0.78
ESSLLI	0.32	0.93	0.72	0.32	0.98	0.72
MITCHELL	0.26	0.97	0.68	0.27	0.97	0.69

Semantic clustering: summary of performance (purity)

Semantic clustering: parameters and explained variance

Feature ablation (model R^2 – AP: 82%; BATTIG: 77%; ESSLLI 58%; MITCHELL 73%)



Semantic clustering: interactions

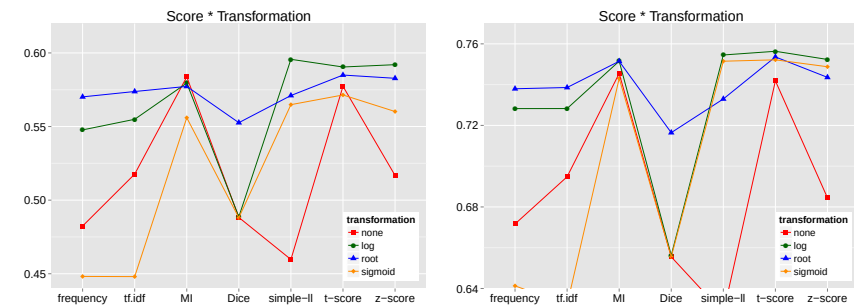
Reduced setting ($R^2 > 0.5$)

Interaction	Df	AP	BATTIG	ESSLLI	MITCHELL
score:transformation	18	7.10	7.95	7.56	11.42
window:metric	4	2.22	1.26	2.97	2.72
metric:n.dim	4	3.29	3.16	2.03	0.58
metric:dim.skip	2	2.25	1.54	2.77	0.86
window:transformation	12	2.00	2.95	0.88	2.66
corpus:metric	2	1.42	2.91	2.79	1.11
corpus>window	8	2.36	1.18	1.49	1.23
score:dim.skip	12	0.56	1.15	0.99	1.39
window:score	24	0.74	0.77	0.54	0.65

Clustering datasets: interactions, R^2

Semantic clustering: Score, Transformation

Partial effect displays (Fox 2003)

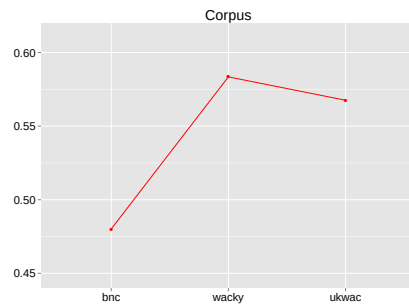


Almuhareb & Poesio

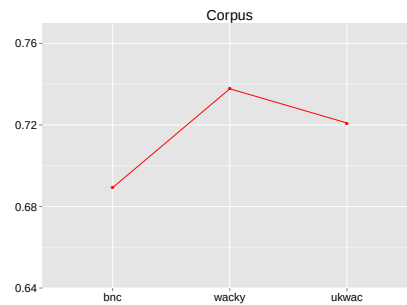
ESSLLI 2008

Semantic clustering: Corpus

Partial effect displays (Fox 2003)



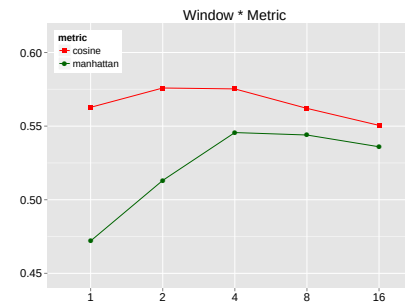
Almuhareb & Poesio



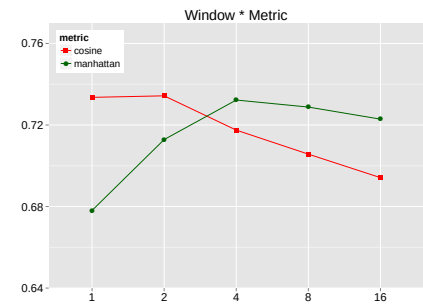
ESSLLI 2008

Semantic clustering: Window Size, Metric

Partial effect displays (Fox 2003)



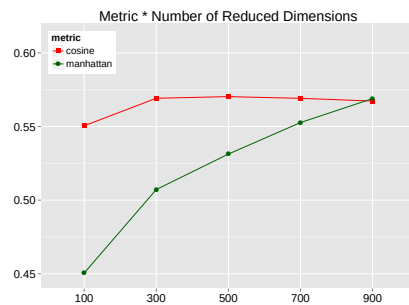
Almuhareb & Poesio



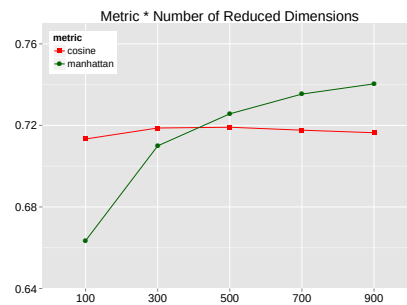
ESSLLI 2008

Semantic clustering: Metric, Number of Latent Dimensions

Partial effect displays (Fox 2003)



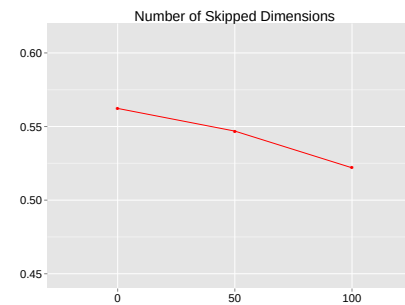
Almuhareb & Poesio



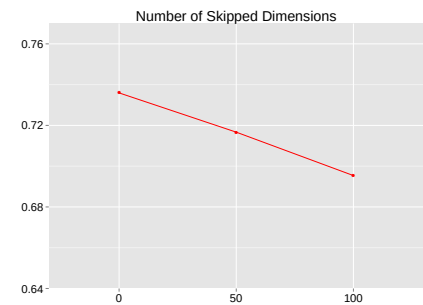
ESSLLI 2008

Semantic clustering: Number of Skipped Dimensions

Partial effect displays (Fox 2003)



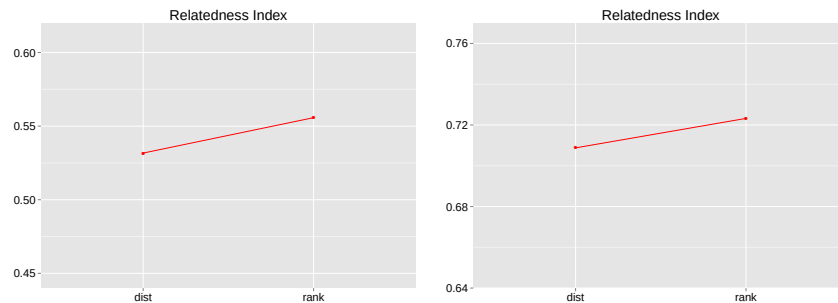
Almuhareb & Poesio



ESSLLI 2008

Semantic clustering: Relatedness Index

Partial effect displays (Fox 2003)

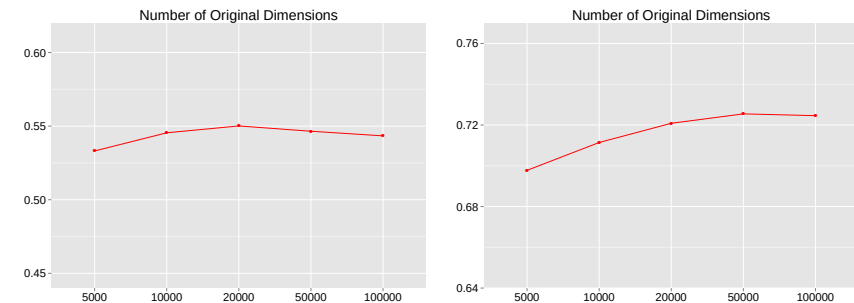


Almuhareb & Poesio

ESSLI 2008

Semantic clustering: Number of Feature Dimensions

Partial effect displays (Fox 2003)



Almuhareb & Poesio

ESSLI 2008

Summing up: Semantic Clustering

Clustering: best setting

- ▶ Corpus: wacky
- ▶ Window: undirected, 4 words
- ▶ Feature selection: top 50000 dimensions, based on frequency
- ▶ Score * Transformation: simple-ll * log (or t-score * log)
- ▶ Dimensionality Reduction: 300/500 latent dimensions, no skipping necessary
- ▶ Distance Metric: cosine
- ▶ Index of Distributional Relatedness: neighbor rank

Outline

Introduction

Tasks & datasets

Evaluation of attributional similarity

Multiple choice

Prediction of similarity ratings

Noun categorisation

Analogy

Methodology for evaluation of DSM parameters

An example (Bullinaria & Levy 2007, 2012)

Interpreting DSM performance with linear regression

Appendix: A large-scale evaluation study

Tasks & parameters

Evaluation results

Summary & conclusion

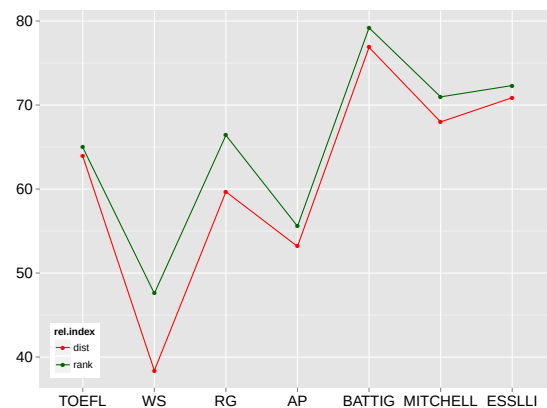
Does our evaluation methodology work?

1. What are the most explanatory parameters?
2. By inspecting the effect plots, we identified best settings for every dataset: what is the performance of such best settings? Are they close to the best runs in the experiment?
3. Is it possible to identify a general best setting that performs reasonably well across all tasks?

Summary: parameters

- ▶ Parameters with strong effect on DSM performance and homogeneous behavior across tasks and datasets
 - ▶ score
 - ▶ transformation
 - ▶ distance metric
- ▶ Parameters with strong effect on DSM performance, but differences across tasks
 - ▶ dimensionality reduction parameters
 - ▶ window
 - ▶ corpus (to a lesser extent)
- ▶ A less crucial parameter with homogeneous behavior
 - ▶ number of context dimensions
- ▶ Parameters that have no or little effect on DSM performance
 - ▶ criterion for context selection
 - ▶ direction of the context window

How about the index of distributional relatedness?



Best settings and their performance

dataset	corpus	w	o.dim	sc	tr	m	rel.ind	n.dim	d.sk	best.s	best.m
TOEFL	ukwac	2	5k	s-II	log	cos	rank	900	100	92.5	98.75
WS	wacky	4	50k	s-II	log	cos	rank	300	50	0.67	0.73
RG	wacky	4	50k	s-II	log	cos	rank	300	50	0.86	0.89
AP	wacky	4	20k	s-II	log	cos	rank	300	0	0.69	0.76
BATTIG	wacky	8	50k	s-II	log	cos	rank	500	0	0.98	0.99
ESSLI	wacky	2	20k	t-sc	log	cos	rank	300	0	0.77	0.98
MITCHELL	wacky	4	50k	s-II	log	cos	rank	500	0	0.88	0.97

Best settings for each dataset

w = window size, o.dim = number of feature dimensions, sc = scoring function, tr = transformation, m = metric, d.sk = number of skipped dimensions, best.s = performance of best setting for this dataset, best.m = performance of best run for this dataset

General settings

task	corpus	w	o.dim	sc	tr.	m	rel.ind	n.dim	d.sk
TOEFL	ukwac	2	5k	s-II	log	cos	rank	900	100
Rating	wacky	4	50k	s-II	log	cos	rank	300	50
Clustering	wacky	4	50k	s-II	log	cos	rank	500	0
General	wacky	4	50k	s-II	log	cos	rank	500	50

General best settings

Task	TOEFL	RATINGS	CLUSTERING	GENERAL	SoA
TOEFL	92.5	85.0	75.0	90.0	100.0
RG	0.85	0.86	0.84	0.87	0.86
WS	0.60	0.67	0.64	0.68	0.81
AP402	0.60	0.66	0.67	0.67	0.79
BATTIG	0.85	0.91	0.98	0.90	0.96
ESSLI	0.70	0.77	0.80	0.77	0.91
MITCHELL	0.73	0.83	0.88	0.83	0.94

General best settings – Performance

References I

- Almuhareb, Abdulrahman (2006). *Attributes in Lexical Acquisition*. Ph.D. thesis, University of Essex.
- Baroni, Marco and Lenci, Alessandro (2010). Distributional Memory: A general framework for corpus-based semantics. *Computational Linguistics*, **36**(4), 673–712.
- Baroni, Marco and Lenci, Alessandro (2011). How we BLESSed distributional semantic evaluation. In *Proceedings of the GEMS 2011 Workshop on GEometrical Models of Natural Language Semantics*, pages 1–10, Edinburgh, UK.
- Baroni, Marco; Dinu, Georgiana; Kruszewski, Germán (2014). Don't count, predict! A systematic comparison of context-counting vs. context-predicting semantic vectors. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics*, pages 238–247, Baltimore, MD.
- Bruni, Elia; Tran, Nam Khanh; Baroni, Marco (2014). Multimodal distributional semantics. *Journal of Artificial Intelligence Research*, **49**, 1–47.
- Budanitsky, Alexander and Hirst, Graeme (2006). Evaluating WordNet-based measures of lexical semantic relatedness. *Computational Linguistics*, **32**(1), 13–47.
- Bullinaria, John A. and Levy, Joseph P. (2007). Extracting semantic representations from word co-occurrence statistics: A computational study. *Behavior Research Methods*, **39**(3), 510–526.

References II

- Bullinaria, John A. and Levy, Joseph P. (2012). Extracting semantic representations from word co-occurrence statistics: Stop-lists, stemming and SVD. *Behavior Research Methods*, **44**(3), 890–907.
- Caron, John (2001). Experiments with LSA scoring: Optimal rank and basis. In M. W. Berry (ed.), *Computational Information Retrieval*, pages 157–169. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA.
- Evert, Stefan (2008). Corpora and collocations. In A. Lüdeling and M. Kytö (eds.), *Corpus Linguistics. An International Handbook*, chapter 58, pages 1212–1248. Mouton de Gruyter, Berlin, New York.
- Finkelstein, Lev; Gabrilovich, Evgeniy; Matias, Yossi; Rivlin, Ehud; Solan, Zach; Wolfman, Gadi; Ruppín, Eytan (2002). Placing search in context: The concept revisited. *ACM Transactions on Information Systems*, **20**(1), 116–131.
- Fox, John (2003). Effect displays in r for generalised linear models. *Journal of Statistical Software*, **8**(15), 1–27.
- Gladkova, Anna; Drozd, Aleksandr; Matsuoka, Satoshi (2016). Analogy-based detection of morphological and semantic relations with word embeddings: what works and what doesn't. In *Proceedings of the NAACL Student Research Workshop*, pages 8–15, San Diego, California.

References III

- Hassan, Samer and Mihalcea, Rada (2011). Semantic relatedness using salient semantic analysis. In *Proceedings of the Twenty-fifth AAAI Conference on Artificial Intelligence*.
- Herdağdelen, Amaç; Erk, Katrin; Baroni, Marco (2009). Measuring semantic relatedness with vector space models and random walks. In *Proceedings of the 2009 Workshop on Graph-based Methods for Natural Language Processing (TextGraphs-4)*, pages 50–53, Suntec, Singapore.
- Hill, Felix; Reichart, Roi; Korhonen, Anna (2015). SimLex-999: Evaluating semantic models with (genuine) similarity estimation. *Computational Linguistics*, **41**(4), 665–695.
- Hutchison, Keith A.; Balota, David A.; Neely, James H.; Cortese, Michael J.; Cohen-Shikora, Emily R.; Tse, Chi-Shing; Yap, Melvin J.; Bengson, Jesse J.; Niemeyer, Dale; Buchanan, Erin (2013). The semantic priming project. *Behavior Research Methods*, **45**(4), 1099–1114.
- Karypis, George (2003). CLUTO: A clustering toolkit. Technical Report 02-017, University of Minnesota, Department of Computer Science, Minneapolis, MN.
- Kaufman, Leonard and Rousseeuw, Peter J. (1990). *Finding Groups in Data: An Introduction to Cluster Analysis*. John Wiley and Sons, New York.

References IV

- Kiela, Douwe and Clark, Stephen (2014). A systematic study of semantic vector space model parameters. In *Proceedings of the 2nd Workshop on Continuous Vector Space Models and their Compositionality (CVSC)*, pages 21–30, Gothenburg, Sweden.
- Landauer, Thomas K. and Dumais, Susan T. (1997). A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction and representation of knowledge. *Psychological Review*, **104**(2), 211–240.
- Lapesa, Gabriella and Evert, Stefan (2014). A large scale evaluation of distributional semantic models: Parameters, interactions and model selection. *Transactions of the Association for Computational Linguistics*, **2**, 531–545.
- Lapesa, Gabriella and Evert, Stefan (2017). Large-scale evaluation of dependency-based DSMs: Are they worth the effort? In *Proceedings of the 15th Annual Meeting of the European Association for Computational Linguistics (EACL 2017): Volume 2, Short Papers*, pages 394–400, Valencia, Spain.
- Levy, Omer; Goldberg, Yoav; Dagan, Ido (2015). Improving distributional similarity with lessons learned from word embeddings. *Transactions of the Association for Computational Linguistics*, **3**, 211–225.

References VI

- Polajnar, Tamara and Clark, Stephen (2014). Improving distributional semantic vectors through context selection and normalisation. In *Proceedings of the 14th Conference of the European Chapter of the Association for Computational Linguistics*, pages 230–238, Gothenburg, Sweden.
- Rogers, Anna; Hosur Ananthakrishna, Shashwath; Rumshisky, Anna (2018). What's in your embedding, and how it predicts task performance. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2690–2703, Santa Fe, NM. Association for Computational Linguistics.
- Rubenstein, Herbert and Goodenough, John B. (1965). Contextual correlates of synonymy. *Communications of the ACM*, **8**(10), 627–633.
- Sahlgren, Magnus (2008). The distributional hypothesis. *Italian Journal of Linguistics*, **20**(1), 33–53.
- Santus, Enrico; Gladkova, Anna; Evert, Stefan; Lenci, Alessandro (2016). The CogALex-V shared task on the corpus-based identification of semantic relations. In *Proceedings of the 5th Workshop on Cognitive Aspects of the Lexicon (CogALex-V)*, pages 69–79, Osaka, Japan.
- Turney, Peter D. (2006). Similarity of semantic relations. *Computational Linguistics*, **32**(3), 379–416.

References V

- Mikolov, Tomas; Sutskever, Ilya; Chen, Kai; Corrado, Greg S.; Dean, Jeff (2013a). Distributed representations of words and phrases and their compositionality. In C. J. C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Q. Weinberger (eds.), *Proceedings of Advances in Neural Information Processing Systems 26 (NIPS 2013)*, pages 3111–3119. Curran Associates, Inc.
- Mikolov, Tomas; Chen, Kai; Corrado, Greg; Dean, Jeffrey (2013b). Efficient estimation of word representations in vector space. In *Workshop Proceedings of the International Conference on Learning Representations 2013*.
- Mikolov, Tomas; Grave, Edouard; Bojanowski, Piotr; Puhrsch, Christian; Joulin, Armand (2018). Advances in pre-training distributed word representations. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, pages 52–55, Miyazaki, Japan.
- Mitchell, Tom M.; Shinkareva, Svetlana V.; Carlson, Andrew; Chang, Kai-Min; Malave, Vicente L.; Mason, Robert A.; Just, Marcel Adam (2008). Predicting human brain activity associated with the meanings of nouns. *Science*, **320**, 1191–1195.
- Padó, Sebastian and Lapata, Mirella (2007). Dependency-based construction of semantic space models. *Computational Linguistics*, **33**(2), 161–199.
- Pennington, Jeffrey; Socher, Richard; Manning, Christopher D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP 2014*.

References VII

- Van Overschelde, James; Rawson, Katherine; Dunlosky, John (2004). Category norms: An updated and expanded version of the Battig and Montague (1969) norms. *Journal of Memory and Language*, **50**, 289–335.